Java Notes: JDK, JRE, and JVM

1. What is JDK (Java Development Kit)?

JDK stands for Java Development Kit. It is used to develop and compile Java programs. It contains the Java compiler (javac), JRE, JVM, and development tools. Functions: Write Java programsCompile Java codeDebug and test applicationsIncludes JRE and JVM

2. What is JVM (Java Virtual Machine)?

JVM executes Java bytecode by converting it into machine code. It makes Java platform independent, while the JVM implementation itself is platform dependent. Functions: Executes bytecodeConverts bytecode to machine codeManages memoryGarbage collection

3. What is JRE (Java Runtime Environment)?

JRE provides the runtime environment to run Java applications. It contains the JVM, Java libraries, and supporting files. Functions: Runs Java programsLoads Java classesProvides Java librariesContains JVM

Relationship

JDK = JRE + Development Tools
JRE = JVM + Java Libraries

Execution Process

Java Source Code (.java) → JDK Compiler (javac) → Bytecode (.class) → JVM → Machine Code → Output

Differences

JDK: Develop, compile, and run.
JRE: Run Java programs.
JVM: Execute Java bytecode.

Interview Points

1. JDK is used for development.
2. JVM executes bytecode.
3. JRE provides the runtime environment.
4. Java is platform independent because bytecode runs on any compatible JVM.
5. Need JDK to compile; JRE is enough to run compiled programs.

Memory Trick

JDK = Develop + Compile + Run
JRE = Run Java Programs
JVM = Execute Bytecode